

## UBT521: AI IN BIOTECHNOLOGY

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**Course Objectives:** Introduce students to Artificial Intelligence applications in biotechnology. Provide understanding of AI-driven genomic, proteomic, and clinical data analysis. Familiarize students with AI tools used in drug discovery, precision medicine, agriculture, and environmental biotechnology. Develop skills to analyze large-scale biological datasets using AI techniques.

### Course Learning Outcomes

After completing the course, students will be able to:

1. Understand fundamental AI concepts and tools used in biotechnology.
2. Apply AI methods to genomic and proteomic data analysis.
3. Interpret AI-based predictions in drug discovery and personalized medicine.
4. Evaluate AI applications in agriculture, environmental biotechnology, and food biotechnology.

### Syllabus

**Unit 1: Introduction to Artificial Intelligence in Biotechnology:** Overview of Artificial Intelligence and Machine Learning, AI software and platforms used in biotechnology; Applications of AI in life sciences, Introduction to biological datasets.

**Unit 2: Genomics and AI:** Basics of genomics, AI in genome sequencing and genome assembly, AI-based genomic data analysis, Pattern recognition in genomic datasets; Tools: DeepVariant – Deep learning tool for accurate SNP and indel detection from sequencing data; AlphaMissense – Predicts effects of missense mutations; Enformer – Predicts gene expression directly from DNA sequences; GATK – Widely used platform for genomic variant discovery and analysis; DeepSEA – Predicts chromatin effects of sequence variants.

**Unit 3: Proteomics and AI:** Basics of proteomics, AI in protein structure prediction, Protein-protein interaction prediction, AI-based functional annotation of proteins; Tools: AlphaFold – AI system for predicting protein 3D structures with high accuracy; RoseTTAFold – Predicts protein structures and interactions; DeepMind AlphaFold Database – Repository of predicted protein structures; DeepInteract – Predicts protein interaction networks.

**Unit 4: AI Applications in Drug Discovery, Precision Medicine, and Agricultural Biotechnology:** Overview of the role of Artificial Intelligence in modern biotechnology, including applications in drug discovery, healthcare, and agriculture. Introduction to the drug discovery pipeline and AI-assisted approaches for drug target prediction, virtual screening, and drug design. AI applications in healthcare include disease diagnosis, predictive modeling, clinical decision support systems, and

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epidemiological analysis. Use of AI in crop genomics, crop improvement, pest resistance, and crop disease prediction. Tools: DeepChem – Platform for molecular modeling and drug discovery; AtomNet – Deep learning platform for predicting drug–target interactions; PlantCV – AI-based analysis of plant images; AgroAI – Platform for crop monitoring and predictive agricultural analytics.

**Practical Component:** Students will be introduced to AI tools and biological databases.

### **Practical Topics**

1. Basic genomic data analysis using AI tools – DeepVariant
2. Basic proteomic data analysis using AI tools – DeepInteract
3. AI-based protein structure prediction – AlphaFold
4. AI tools for virtual screening – AutoDock Vina
5. Predictive modeling using biological datasets – PlantCV, AgroAI
6. Case study analysis using AI tools – Digital twins

### **Text Books**

1. Zhang, Yanqing; Rajapakse, Jagath C.; Zomaya, Albert Y.; Pan, Yi (2008). *Machine Learning in Bioinformatics* John Wiley & Sons (Wiley-Interscience). ISBN: 978-0470116623.
2. Compeau, P., & Pevzner, P. (2015). *Bioinformatics Algorithms: An Active Learning Approach*. La Jolla, CA: Active Learning Publishers.
3. Lesk, A. M. (2019). *Introduction to Bioinformatics* (5th ed.). Oxford: Oxford University Press.

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